

HTML Tags: Anchor, Form, Table, Image, List Tags, Paragraph, Break, Label

1. Introduction to HTML and its structure.

- HTML (HyperText Markup Language) is the standard language used to create webpages. It defines the structure of web content using tags and elements.
- `<html>`
 `<head>`
 `<title>My Page</title>`
 `</head>`
 `<body>`
 Content Here
 `</body>`
 `</html>`

2. Explanation of key tags:

- `<a>`: Anchor tag for hyperlinks.
 - Anchor tag is used to create hyperlinks between webpages.
 - `<a href="https://google.com">Google</a>`
 - Navigate between pages
 - Open websites
 - Create downloadable links
- `<form>`: Form tag for user input.
 - The form tag is used to collect user input.
 - `<form> <input type="text"> </form>`
 - Login forms , Registration forms , Feedback forms

- `<table>`: Table tag for data representation.
- Table tag is used to display data in rows and columns.
- `<table border="1">`
- `<tr>`
- `<td>Name</td>`
- `</tr>`
- `</table>`
- `<table>` → Creates table
- `<tr>` → Table row
- `<td>` → Table data
- `<th>` → Table heading

- `<img>`: Image tag for embedding images.
- Used to display images on webpage.
- ``
- `src` → Image path
- `alt` → Alternative text

- List tags: `<ul>`, `<ol>`, and `<li>`
- Used to create lists.
- `<ol>`
- `<li>Java</li>`
- `<li>HTML</li>`
- `</ol>`
- Displays bullet list.
- `<ul>`

- `<li>Apple</li>`
- `<li>Mango</li>`
- `</ul>`
- Represents list items.

- `<p>`: Paragraph tag.
- Used to write paragraphs.
- `<p>This is paragraph.</p>`

- `<br>`: Line break.
- Used to break line.
- `Hello<br>World`

- `<label>`: Label for form inputs.
- Label tag defines label for form elements.
- `<label>Name</label>`
- `<input type="text">`
- Improves accessibility , User-friendly forms

CSS: Inline CSS, Internal CSS, External CSS

1. Overview of CSS and its importance in web design.
 - CSS (Cascading Style Sheets) is used to design and style webpages.
 - It controls:

- Colors , Fonts , Layout , Spacing , Animations
- Advantages:
 - Makes webpages attractive ,Separates design from HTML ,Reusable styles ,Faster webpage maintenance ,Responsive web design support

2. Types of CSS:

- Inline CSS: Directly in HTML elements.
 - Applied directly inside HTML element.
 - `<p style="color:red;"> Hello </p>`
 - Advantages:
 - Quick styling
 - Disadvantages:
 - Difficult to manage
- Internal CSS: Inside a `<style>` tag in the head section.
 - Written inside `<style>` tag.
 - `<style>`
 - `p{`
 - `color:blue;`
 - `}`
 - `</style>`
- External CSS: Linked to an external file.
 - CSS stored in separate .css file.
 - `<link rel="stylesheet" href="style.css">`
 - Advantages:
 - Reusable , Easy maintenance

CSS: Margin and Padding

3. Definition and difference between margin and padding.

4. How margins create space outside the element and padding creates space inside

➤ Margin creates space outside the element border.

➤ `div{ margin:20px;}`

➤ Padding creates space inside the element border.

➤ `div{ padding:20px; }`

➤ Margin	Padding	
➤ -----	-----	
➤ Outside spacing	Inside spacing	
➤ Transparent	Background color visible	
➤ Controls distance between elements Controls content spacing		

CSS: Pseudo-Class

1. Introduction to CSS pseudo-classes like :hover, :focus, :active, etc.

2. Use of pseudo-classes to style elements based on their state.

➤ Pseudo-classes are used to define special states of elements.

➤ Examples:

➤ :hover , :focus , :active

➤ `a:hover{ color:red; }`

- When mouse moves over link, color changes.
- Uses of Pseudo-Classes
- Button effects ,Form focus styling ,Hover animations ,Interactive webpages

CSS: ID and Class Selectors

1. Difference between id and class in CSS.
2. Usage scenarios for id (unique) and class (reusable).
 - ID is unique for one element.
 - `#header{ color:red; }`
 - Class is reusable for multiple elements.
 - `.box{ background:yellow; }`
 - | ID | Class |
 - | ----- | ----- |
 - | Unique | Reusable |
 - | Uses # symbol | Uses . symbol |
 - | Applied once | Applied many times |

Introduction to Client-Server Architecture

1. Overview of client-server architecture.
2. Difference between client-side and server-side processing.
3. Roles of a client, server, and communication protocols.
 - Client-server architecture is a model where:

- Client sends request ,Server processes request , Server sends response
- Client-Side Processing
- Performed on user browser.
- Examples:
- HTML
- CSS , JavaScript
- Server-Side Processing
- Performed on server.
- Examples:
- Java ,PHP , Python
- Roles
- Client
- Requests services.
- Server
- Provides services.
- Protocol
- Communication rules like HTTP.

HTTP Protocol Overview with Request and Response Headers

1. Introduction to the HTTP protocol and its role in web communication.
2. Explanation of HTTP request and response headers.
 - Introduction to HTTP
 - HTTP (HyperText Transfer Protocol) is used for communication between browser and server.
 - HTTP Request Header

- Contains:
- Browser information
- Request type
- Cookies
- Example:
- GET /index.html HTTP/1.1
- HTTP Response Header
- Contains:
- Status code
- Content type
- Server information
- Example:
- HTTP/1.1 200 OK

J2EE Architecture Overview

1. Introduction to J2EE and its multi-tier architecture.
2. Role of web containers, application servers, and database servers.
 - Introduction
 - J2EE (Java 2 Enterprise Edition) is used to develop enterprise applications.
 - Layers of J2EE
 - 1. Presentation Layer
 - User interface.
 - 2. Business Logic Layer
 - Processes application logic.
 - 3. Data Layer

- Database operations.
- Components
- Web Container
- Application Server
- Database Server

Web Component Development in Java (CGI Programming)

1. Introduction to CGI (Common Gateway Interface).
2. Process, advantages, and disadvantages of CGI programming.
 - Introduction to CGI
 - CGI (Common Gateway Interface) allows web server to interact with external programs.
 -
 - Advantages
 - Simple
 - Language independent
 -
 - Disadvantages
 - Slow
 - Creates new process for every request
 - High memory usage

Servlet Programming: Introduction, Advantages, and Disadvantages

1. Introduction to servlets and how they work.
2. Advantages and disadvantages compared to other web technologies.

- Introduction to Servlet
- Servlet is a Java program that runs on server and handles client requests.
- Advantages of Servlet
 - Fast
 - Platform independent
 - Secure
 - Reusable
- Disadvantages
 - Complex compared to HTML
 - Requires server configuration

Servlet Versions, Types of Servlets

1. History of servlet versions.
2. Types of servlets: Generic and HTTP servlets.
 - GenericServlet
 - Protocol independent servlet.
 - HttpServlet
 - Used for HTTP protocol.
 - Supports:
 - GET
 - POST
 - PUT
 - DELETE

Difference between HTTP Servlet and Generic Servlet

1. Detailed comparison between HttpServlet and GenericServlet.

GenericServlet	HttpServlet	
-----	-----	
Protocol independent	HTTP specific	
Uses service()	Uses doGet(), doPost()	
Less features	More web features	

Servlet Life Cycle

Explanation of the servlet life cycle: init(), service(), and destroy() methods.

Methods

1. init()

Called once during servlet initialization.

2. service()

Handles requests.

3. destroy()

Called before servlet removal.

Creating Servlets and Servlet Entry in web.xml

How to create servlets and configure them using web.xml.

ServletConfig is used to get servlet initialization parameters.

Methods

`getInitParameter()`

`getServletName()`

`getServletContext()`

RequestDispatcher

Definition

Used to forward request from one resource to another.

Methods

`forward()`

Transfers request.

`include()`

Includes another resource output.

ServletContext

Definition

ServletContext shares data between servlets.

Advantages

Application-wide data sharing

Resource management

Java Filters

Definition

Filters are used to filter requests and responses.

Uses

Validation

Authentication

Logging

JSP Basics

JSP

JSP (Java Server Pages) is used to create dynamic webpages.

Components

JSTL

Custom Tags

Scriptlets

Implicit Objects

Session Management and Cookies

Session Management

Used to track users in web applications.

Techniques

Cookies

Hidden fields

URL rewriting
HttpSession

Cookies

Small files stored in browser to maintain user information.